CLINICAL NEPHROLOGY & COMPREHENSIVE PATHOLOGY REFERENCE

Kidney Disease Study Guide

Integrative analysis of renal pathologies across Allopathy, Ayurveda, Homeopathy, and Unani medical systems.

1. Diagnostic Biomarkers & Staging Cutoffs

Standard lab parameters utilized in clinical evaluation and diagnosis across key kidney disease categories:

Condition	Primary Laboratory & Imaging Biomarkers	Key Diagnostic Cutoffs
Chronic Kidney Disease (CKD)	eGFR, Serum Creatinine, Urine Albumin-to-Creatinine Ratio (UACR)	eGFR < 60 mL/min/1.73m² for >3 months; UACR > 30 mg/g
Nephrotic Syndrome	24-hr Urine Protein, Serum Albumin, Lipid Panel	Heavy Proteinuria (>3.5 g/24hr), Hypoalbuminemia (<3.0 g/dL)
Glomerulonephritis	Urine Sediment (Dysmorphic RBCs & RBC Casts), Serum C3/C4	Active urinary sediment; Nephritic pattern with dysmorphic cells
Lupus Nephritis	Anti-dsDNA, Serum C3/C4 complement levels, Renal Biopsy	ISN/RPS Classes I–VI biopsy staging; Class IV wire-loop lesions
Diabetic Nephropathy	Urine Albumin Excretion (UACR), HbA1c, Serum Creatinine	Microalbuminuria (30–300 mg/day) progressing to >300 mg/day
Acute Kidney Injury (AKI)	Serum Creatinine, Blood Urea Nitrogen (BUN), Urine Output	KDIGO: Cr rise ≥ 0.3 mg/dL within 48h or ≥ 1.5× baseline
Renal Tubular Acidosis (RTA)	Serum Bicarbonate, Serum K^+, Urine pH, Urine Anion Gap	Normal Anion Gap Acidosis; Urine pH > 5.5 in Type 1 RTA
Myeloma Kidney	Serum Free Light Chains (κ/λ ratio), Urine Protein Electrophoresis	Bence-Jones proteinuria; FLC ratio > 100 with light chain casts
Kidney Stones	24-hr Urine Chemistry (Ca, Oxalate, Uric acid), Non-contrast CT	Hypercalciuria, Hypocitraturia; CT Hounsfield Unit density

2. Glomerular & Podocyte Pathology Overview

Nephrological disorders target the internal filtering sheets (glomeruli and podocytes). Mainstream treatment centers on immunosuppression and RAAS blockade to halt cellular scarring, while traditional medicine utilizes Pitta-cooling, blood-purifying, and drainage therapies.